

Project 20

St. George Hospital Management System

Objectives

The client requires you to craft a top-notch centralized mobile healthcare management system project for their healthcare facilities, owning multiple branches of super-specialty hospitals in the different states of Australia. The client's primary aim was to establish a single healthcare management system, connecting remote branches and handling their core operations from the headquarters.

Before contacting your company, the client used an outdated healthcare management system comprising various vulnerable loopholes, attracting cyberattackers to breach patients' personal and health records.

You need to insightfully examine the client requirements to fabricate healthcare management software solution aligning with hospitals and government policies and fulfilling their dynamic business needs.

Existing System:

Difficulty in Sharing Data

Client had multiple branches of their hospital at distant locations, and it was becoming complex to share files over the mail. Sometimes the mail is received in the spam folder, and managing data becomes more complicated. For storing any file, the authorized person first has to download it on the system and then store it in the patient's folder.

The main business objective to mitigate this challenge was to develop a central database management system, which any legitimate person could access in the hospital's network.

Streamlining Patient's Appointment Process

With previous hospital staff management software, users were calling at the hospital's reception for making their appointment and for resolving their

queries. The hospital staff was constantly under pressure to pick up the call and update the doctor about their schedule simultaneously.

With the new healthcare management system, the hospital wants to provide an online platform for patients to book their appointments, which should be reflected in the doctor's account. Also, patients can view that and book for some other day once the slots are full.

Managing Hospital Data

Client was facing difficulties in managing their data in physical form. Whenever a patient arrives at the hospital, the staff makes a new file and updates data in the hospital's software. With this, they faced space shortage in storing files and other documents.

The goal here was to digitize the storage of necessary documents and organize them in a meaningful manner. In addition, client requires shortening the time to retrieve previous health record data and eliminating the use of paper on the premises.

Data Security

With the outdated system, all the sensitive information, such as patients' health and personal data, was stored in a non-encrypted format. Everyone using the system had read and write permissions to it. Also, it was not aligning with the latest HIPPA and NIST policies.

Client needed to configure a security mechanism to allow only authorized users to access the data. Moreover, the data must be safe from cyber-attacks and maintain its integrity, confidentiality, and availability.

Accessibility

The system was only available to staff members for inputting data and sharing it with the doctors and lab associates through the mail using the integrated API.

The client wants to make the application accessible to every stakeholder for automating the core operations and connecting the different departments in multiple branches through a secure transmission channel.

Proposed Solution:

The proposed system consists of four panels: Doctor, Patient, Hospital or clinic and Admin. The users will first have to download the application and install it in their mobile devices. Once installed, this application will remain into the device permanently until the user deletes it or uninstalls it. The patient will have to register into the application for the first time. On registering, the patient will receive a username and password. The patient can use this username and password for logging into the app each time he/she uses it. After logging in, the patient will have to select a filtration type. The filtration is done on two bases: Gender wise and Specialty wise. After selecting the filtration type, the doctors list will be displayed. The patient can select any particular doctor and view his profile. And patient give reviews on doctor profile. Also the patient can view the doctor's profile and look for an appointment. The patient will then send a request for appointment. The doctor can either accept the appointment or reject it. The database will get updated accordingly and the patient will get a confirmation message. The add-on to this system is that the patient will receive a notification 2 hours before the actual appointment. As well as if doctor cancels the appointment patient received a message for appointment cancelation. This will be very useful in case the patient tends to forget the appointment. Also doctor can search patient history by using a unique ID. Adding AES encryption with a public key cryptosystem to the proposed methods is one possibility to eradicate security leaks.

OVERALL DESCRIPTION OF THE PROPOSED SYSTEM

Module Description

This system enables user friendly for Doctor-Patients relationship. Doctor can easily diagnosis patient's problem and find out the solution. Then clinic provide update details about patient's problems.

The share is encrypted using AES encryption with a public key selected by hospital server. The secret image (share 1) is converted to array of bytes using

getImageBytes method. The converted image bytes are encrypted using AES encryption technique. The only request with the public key can be able to pass through the authentication process. After the share is encrypted using AES encryption LSB embedding is applied on the share.

System Features

In the life of the software development, problem analysis provides a base for design and development phase. The problem is analyzed so that sufficient matter is provided to design a new system. Large problems are sub-divided into smaller ones to make them understandable and easy for finding solutions. Same in this project all the tasks are sub-divided and categorized.

System Modules:

1. User Authentication and Role Management

- Handles login, logout, registration and password encryption.
- Enforces role-based access for Admin, Branch Manager, Doctor, Nurse, Receptionist and Patient.
- Includes session management and activity logs.

2. Branch Management Module

- Allows the Admin to add, update or deactivate branches.
- Stores branch-specific information such as address, contact and capacity.
- Enables branch-wise dashboards and statistics for managers.

3. Patient Management Module

- Registers new patients and assigns a global patient ID.
- Stores demographic, medical and contact information.
- Manages patient records, allergies and treatment histories.

4. Appointment and Scheduling Module

- Displays doctors' schedules and available time slots.

- Enables patients or receptionists to book, reschedule or cancel appointments.
- Prevents time-slot conflicts and sends reminders via email/SMS.

5. Doctor and Staff Management Module

- Manages profiles, qualifications, and work schedules of doctors and staff.
- Tracks shift allocation, attendance and departmental assignments.
- Allows doctors to access patient records and update treatment progress.

6. In-Patient (Admission and Ward) Management Module

- Records patient admissions, room/bed assignments and discharge details.
- Tracks daily vitals and nurse observations.
- Links billing and pharmacy data to admitted patients.

7. Pharmacy and Inventory Module

- Maintains a central database of medicines with quantities, expiry and supplier details.
- Issues medicines to patients and deducts stock automatically.
- Generates low-stock and expiry alerts supports purchase orders.

8. Laboratory and Diagnostics Module

- Allows doctors to request pathology or imaging tests.
- Lets lab technicians upload results and attach digital reports.
- Stores branch-wise test histories and sends notifications to patients.

9. Billing and Payments Module

- Generates automatic itemised invoices for consultations, tests and medicines.
- Tracks payment modes (cash, card, online) and status (paid/pending).
- Integrates optional online payment gateways and maintains transaction records.

10. Reports and Analytics Module

- Produces statistical reports: patient volume, revenue, test usage, inventory etc.

- Exports reports in PDF/Excel formats.
- Provides visual dashboards for admins and branch managers.

11.Communication and Notification Module

- Sends automated notifications for appointments, bills and test results.
- Supports SMS/email reminders and broadcast announcements.
- Logs all system messages for audit tracking.

12.Audit and Security Management Module

- Records user activities, login attempts and data modification logs.
- Enables admins to monitor suspicious or unauthorised actions.
- Supports audit-trail generation for compliance purposes.

13.Data Backup and Recovery Module

- Automates daily database backups and stores them securely.
- Provides one-click restore functionality in case of data loss or corruption.
- Ensures business continuity across all branches.

14.System Configuration and Administration Module

- Allows the Admin to configure branch settings, departments and roles.
- Manages system parameters such as appointment duration, notification frequency and backup schedules.
- Provides maintenance utilities for database cleanup and updates.

Functional Requirements

Functional requirements are product features or functions that developers must implement to enable users to accomplish their tasks. So, it's important to make them clear for the stakeholders. Generally, functional requirements describe system behavior under specific conditions. The developers of this system must enhance the performance and efficiency of the system by adding 15 to 20 more functional requirements. Students need to do their own research to find how they can improve the system and which FRs need to be added. The group must need a prior approval from the stakeholders/project

supervisor before finalizing these Functional Requirements. These enhanced FRs must be reflected separately in Final SRS Report after the approval.

User Authentication and Access Control

FR1. The system should allow users to register with valid email, username and password.

FR2. The system should verify login credentials and restrict access to unauthorised users.

FR3. The system should encrypt passwords using secure hashing before storage.

FR4. The system should provide role-based dashboards according to user type (Admin, Doctor etc.).

FR5. The system should allow administrators to add, deactivate or modify user accounts.

Multi-Branch Management

FR6. The system should allow the admin to create and manage multiple hospital branches.

FR7. Each branch should have a unique branch ID and location details. **FR8.** The system should allow the admin to assign staff members to specific branches.

FR9. The system should maintain separate statistics and reports per branch.

FR10. Branch managers should be able to monitor patient admissions, discharges and resources within their branch.

Patient Registration and Records

FR11. The system should allow receptionists to register new patients with demographic details.

FR12. The system should assign a unique patient ID across all branches. **FR13.** The system should allow authorised staff to edit or update patient information.

FR14. The system should store complete medical histories, including past diagnoses and prescriptions.

FR15. The system should allow patients to view their own medical records securely after login.

Appointment and Scheduling

FR16. Patients should be able to request appointments with available doctors.

FR17. The system should display doctor availability by date and time.

FR18. Receptionists should approve, modify, or cancel appointments.

FR19. The system should prevent overlapping appointments for the same doctor.

FR20. The system should send automated confirmation and reminder emails or SMS to patients.

Doctor and Staff Management

FR21. The system should allow admin to add or update doctor and staff profiles (specialisation, contact info, etc.).

FR22. The system should store doctor schedules, including visiting hours and leave dates.

FR23. Doctors should be able to access assigned patient files and update treatment notes.

FR24. The system should allow nurses to record daily observations such as vitals or treatment progress.

FR25. The system should maintain staff attendance and shift allocation records.

Pharmacy and Inventory

FR26. The system should maintain a list of medicines, quantities, expiry dates and suppliers.

FR27. The system should automatically notify pharmacists when stock levels fall below minimum threshold.

FR28. The system should record every medicine issued to patients and link it to their prescriptions.

FR29. The system should generate purchase orders when restocking is required.

FR30. The system should restrict expired medicines from being dispensed.

Laboratory and Diagnostics

FR31. The system should allow doctors to request laboratory or imaging tests for patients.

FR32. Lab technicians should record results, upload scanned reports and link them to patient records.

FR33. The system should notify doctors and patients once test results are available.

FR34. Patients should be able to download or print their lab reports. **FR35.** The system should maintain a searchable database of all laboratory test results per branch.

Billing and Payments

FR36. The system should automatically calculate consultation, test and medication charges.

FR37. The system should generate itemised bills for each patient visit or admission.

FR38. Patients should be able to download or print their invoices from the portal.

FR39. The system should record payment status (paid, pending, cancelled).

FR40. The system should integrate with online payment gateways for electronic transactions.

Reports and Analytics

FR41. The system should generate daily, weekly, and monthly statistics for patients, revenue and appointments.

FR42. The admin should be able to export reports in PDF and Excel formats.

FR43. The system should display graphical dashboards summarising key metrics.

FR44. Branch managers should be able to view comparative performance of departments.

Communication and Notifications

FR45. The system should send automated notifications and reminders (via email/SMS) to staff and patients regarding appointments, reports and billing updates.

FR46. Intelligent Chatbot Assistant

The system should provide an AI-based chatbot that helps patients with common queries such as booking appointments, checking test results, hospital hours or contact information.

(Can use Dialogflow, OpenAI API, or a simple rule-based PHP chatbot engine.)

FR47. Feedback Sentiment Analysis

The system should analyse patient feedback comments using simple sentiment analysis (positive, negative, neutral) to monitor service quality. (Can be done via a basic free sentiment API integration.)

FR48. Automated Report Insights

The system should automatically generate short AI-based summaries of monthly reports (e.g. Patient visits increased by 12% compared to last month). (Can be done using a text-generation or summarisation API.)

Non-Functional Requirements

Performance

NFR1. The system should load each web page within 3 seconds under normal internet conditions.

NFR2. The system should support at least 200 concurrent users without noticeable performance degradation.

NFR3. Database queries should be optimised using indexing to ensure quick retrieval of patient and appointment data.

NFR4. The system should process report generation (e.g. daily statistics) within 10 seconds for datasets under 10,000 records.

Scalability

NFR5. The system architecture should allow seamless addition of new hospital branches without structural database changes.

NFR6. The web application should support deployment across multiple servers if user load increases.

NFR7. The system should maintain consistent performance even as patient and record volume grows annually.

Security

NFR8. All passwords should be stored using bcrypt or equivalent encryption algorithms.

NFR9. The system should use HTTPS (SSL/TLS) for secure data transmission between client and server.

NFR10. Role-based access control should restrict sensitive modules (e.g. billing, medical records) to authorised roles only.

NFR11. The system should automatically log out inactive users after 10 minutes of inactivity.

NFR12. All login attempts, data modifications, and deletions should be logged for auditing.

Usability

NFR13. The user interface should be intuitive, consistent and mobile responsive, compatible with all major browsers.

NFR14. All data-entry forms should include input validation, error messages and tooltips for guidance.

NFR15. The system should support accessibility standards (e.g. ARIA labels, font contrast) for visually impaired users.

Reliability and Availability

NFR16. The system should achieve 99.5% uptime during scheduled operational hours.

NFR17. The system should automatically back up the database every 24 hours and maintain backups for the last 30 days.

NFR18. The system should recover from critical failures (e.g. database crash) within 30 minutes of detection.

Maintainability

NFR19. The source code should follow a Modular MVC architecture to enable easier maintenance and upgrades.

NFR20. System documentation should be maintained for administrators and developers, including installation steps, schema diagrams and update logs.

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.